Lecture 13: Linearity, Thévenin and Norton Equivalents

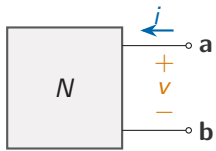
EE201 – Circuit Theory I

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Middle East Technical University

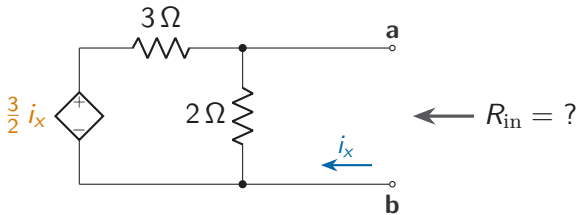
current voltage, node voltage e Δ Y

Review: equivalent one-ports



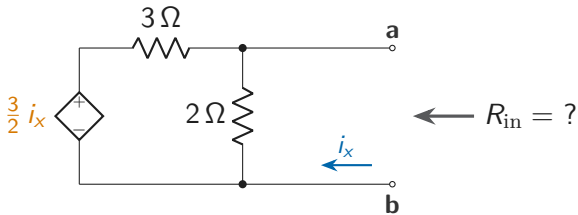
- Equivalent: the same i–v characteristic
- Series, parallel, source transformation

Motivation



- A dependent source inside: series, parallel are not enough

Motivation



- A dependent source inside: series, parallel are not enough
- One equivalent resistor? Yes: the input resistance

Roadmap



Input resistance

1

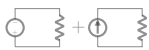
- **a test source**
- **a dependent source**



Linearity

2

- $y = \mathbf{k}\,\mathbf{u}$
- the implication



Superposition

3

- one at a time
- three experiments

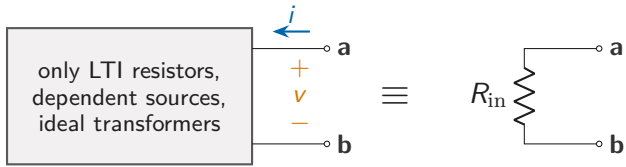


Thévenin, Norton

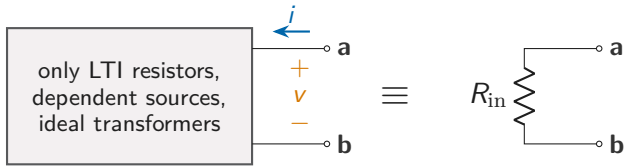
4

- the two theorems
- open, short
- a negative R_{th}
- two examples

One-ports without independent sources

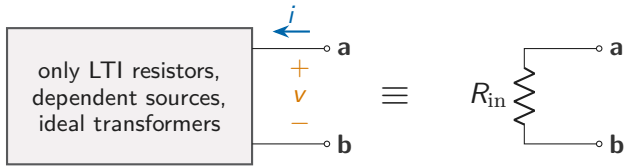


One-ports without independent sources



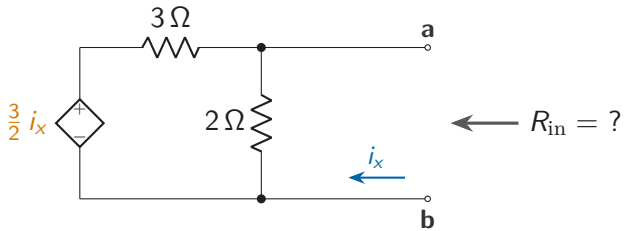
- Controls of the dependent sources: inside the one-port

One-ports without independent sources

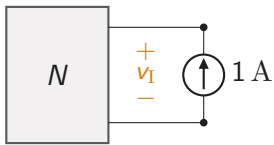


- Controls of the dependent sources: inside the one-port
- $v = R_{\text{in}}\, i$: R_{in} is the input resistance

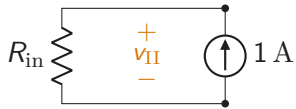
Example 13.1: an input resistance



The idea: a test source

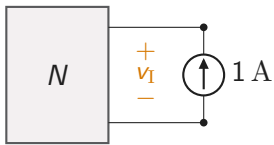


(I)

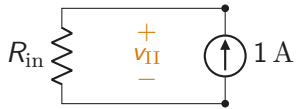


(II)

The idea: a test source



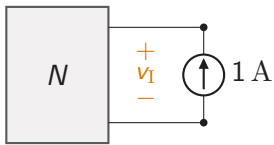
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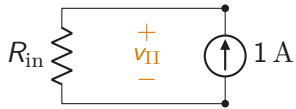
(II)

$$N \equiv R_{\mathrm{in}} \;\Rightarrow\; v_{\mathrm{I}} = v_{\mathrm{II}} = R_{\mathrm{in}} \cdot 1\,\mathrm{A}$$

The idea: a test source



(I)

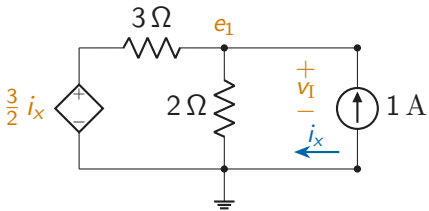


(II)

$$N \equiv R_{\mathrm{in}} \;\Rightarrow\; v_{\mathrm{I}} = v_{\mathrm{II}} = R_{\mathrm{in}} \cdot 1\,\mathrm{A}$$

$$R_{\mathrm{in}} = \frac{v_{\mathrm{I}}}{1\,\mathrm{A}}\text{: solve circuit (I)}$$

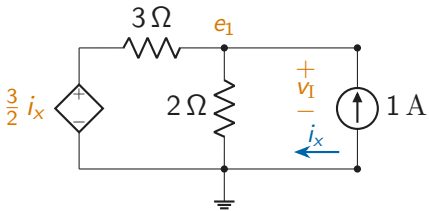
Example 13.1: the solution



Node 1:

$$\frac{e_1 - \frac{3}{2}\, i_x}{3} + \frac{e_1}{2} - 1 = 0$$

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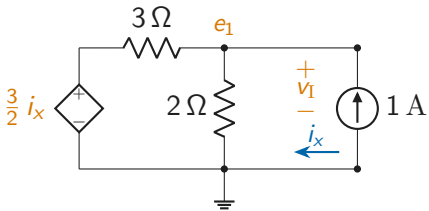


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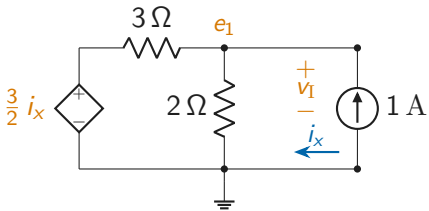
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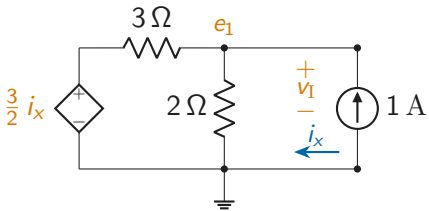
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$$\frac{e_1 + \frac{3}{2}}{3} + \frac{e_1}{2} - 1 = 0 \;\Rightarrow\; e_1 = \tfrac{3}{5}\,\text{V}$$

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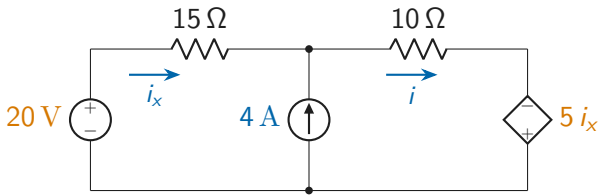
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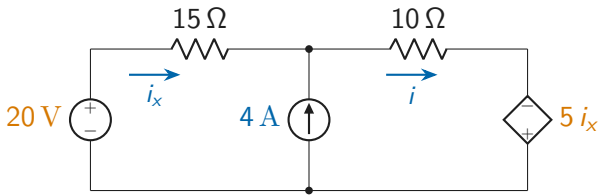
$$R_\mathrm{in} = \frac{v_\mathrm{I}}{1\,\mathrm{A}} = \frac{3}{5}\,\Omega$$

Motivation: many sources



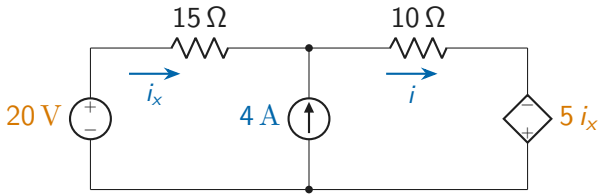
- Two independent sources: the inputs

Motivation: many sources



- Two independent sources: the inputs
- One source at a time: two simpler circuits?

Motivation: many sources

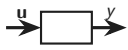


- Two independent sources: the inputs
- One source at a time: two simpler circuits?
- Yes: the circuit is linear

Roadmap



Input resistance



Linearity



Superposition



Thévenin, Norton

✓ — **2** — 3 — 4

a test source ✓

a dependent source ✓

$y = \mathbf{k}\,\mathbf{u}$

the implication

one at a time

three experiments

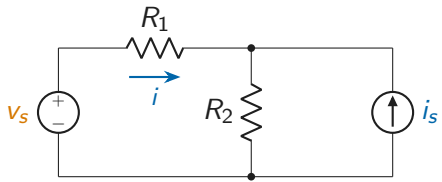
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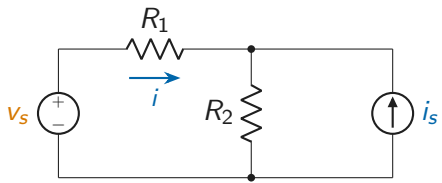
Linearity: inputs and outputs



Input vector: the independent sources

$$\mathbf{u} = \begin{bmatrix} v_s \\ i_s \end{bmatrix}$$

Linearity: inputs and outputs



Input vector: the independent sources

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Output y: any branch voltage or current

e.g. $y = i$

Linearity

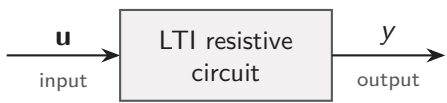
For each output y, there is $\mathbf{k} = [\, k_1 \ \cdots \ k_m \,]$ such that

$$y = \mathbf{k}\,\mathbf{u} = k_1 u_1 + k_2 u_2 + \cdots + k_m u_m$$

Linearity

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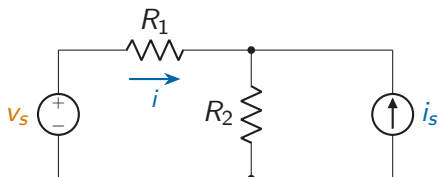
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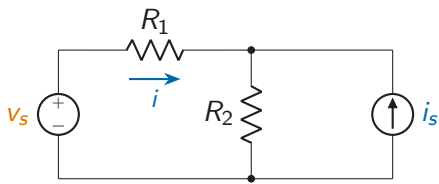
$$y = y(\mathbf{u})$$

Example 13.2: k for $y = i$

$i = k_1 v_s + k_2 i_s$: $k_1, k_2 = ?$



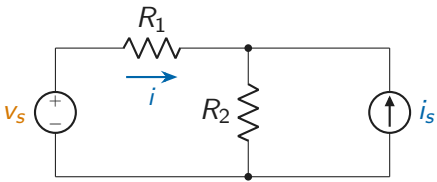
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KVL: $-v_s + R_1 i + R_2(i + i_s) = 0$

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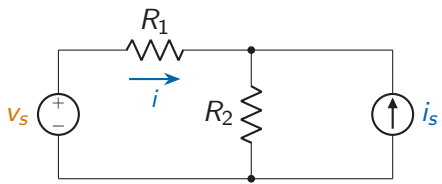


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$$k_1 = \frac{1}{R_1 + R_2}, \qquad k_2 = -\frac{R_2}{R_1 + R_2}$$

Linearity: the implication

$$y(\alpha\,\mathbf{u} + \beta\,\mathbf{w}) = \alpha\,y(\mathbf{u}) + \beta\,y(\mathbf{w})$$

$\mathbf{u}$, $\mathbf{w}$: input vectors; α, β: numbers

Linearity: the implication

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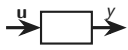
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Superposition: one independent source at a time

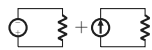
Roadmap



Input resistance



Linearity



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Thévenin, Norton



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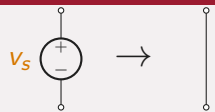
open, short

a negative R_{th}

two examples

Killing a source

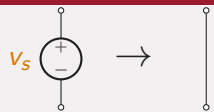
Voltage source



$v_s = 0$: a short circuit

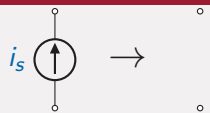
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Voltage source



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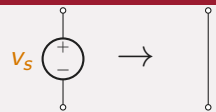
Current source



$i_s = 0$: an open circuit

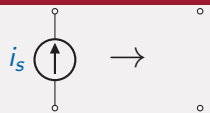
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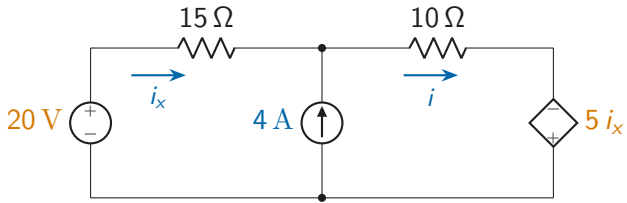
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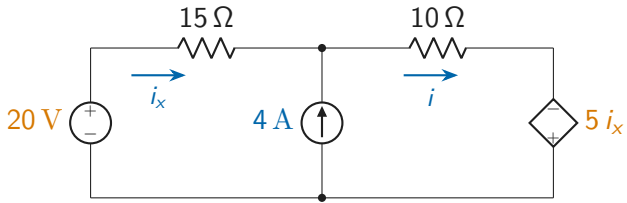
Do not kill dependent sources: they are not inputs

Example 13.3: superposition



Find i, one source at a time

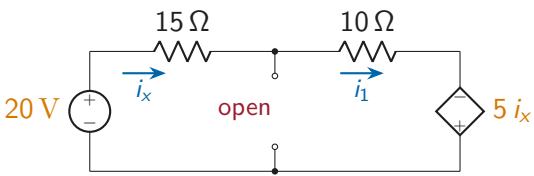
Example 13.3: superposition



Find i, one source at a time

$i = i_1 + i_2$: i_1: only 20 V, i_2: only 4 A

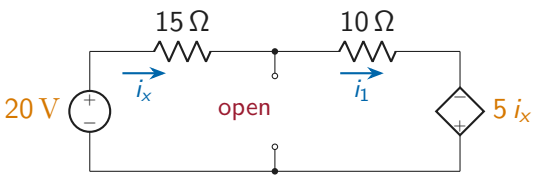
Example 13.3: kill the current source



KVL:

$$-20 + 15\, i_x + 10\, i_1 - 5\, i_x = 0$$

Example 13.3: kill the current source

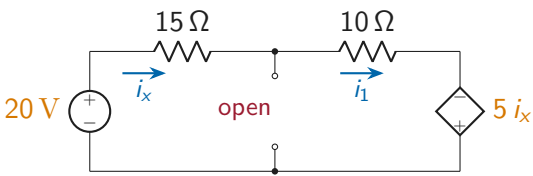


KVL:

$$-20 + 15\, i_x + 10\, i_1 - 5\, i_x = 0$$

$$i_x = ?$$

Example 13.3: kill the current source



KVL:

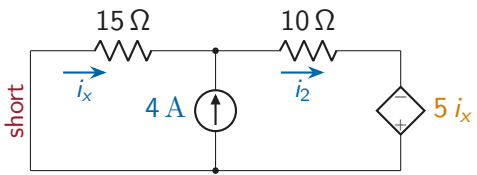
$$-20 + 15\, i_x + 10\, i_1 - 5\, i_x = 0$$

$i_x = ?$

$i_x = i_1 \;\Rightarrow\; 20\, i_1 = 20$

$$\boxed{i_1 = 1\,\text{A}}$$

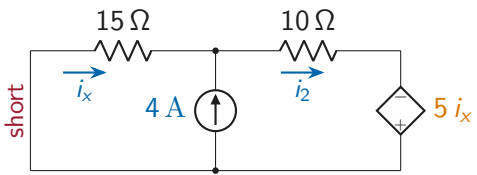
Example 13.3: kill the voltage source



KVL:

$$15\, i_x + 10\, i_2 - 5\, i_x = 0$$

Example 13.3: kill the voltage source

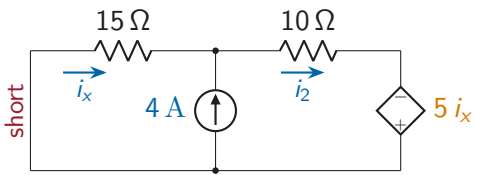


KVL:

$15\,i_x + 10\,i_2 - 5\,i_x = 0$

$i_x = ?$

Example 13.3: kill the voltage source



KVL:

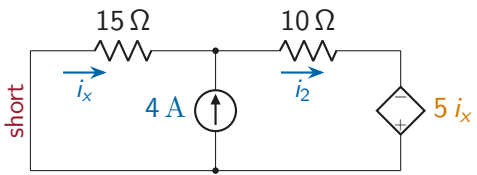
$15\,i_x + 10\,i_2 - 5\,i_x = 0$

$i_x = ?$

$i_x = i_2 - 4 \;\Rightarrow\; 20\,i_2 = 40$

$i_2 = 2\,\text{A}$

Example 13.3: kill the voltage source



KVL:

$15\,i_x + 10\,i_2 - 5\,i_x = 0$

$i_x = ?$

$i_x = i_2 - 4 \;\Rightarrow\; 20\,i_2 = 40$

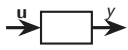
$i_2 = 2\,\mathrm{A}$

$i = i_1 + i_2 = 3\,\mathrm{A}$

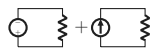
Roadmap



Input resistance



Linearity



Superposition



Thévenin, Norton



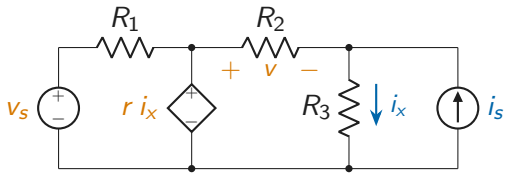
a test source ✓
a dependent source ✓

$y = \mathbf{k}\,\mathbf{u}$ ✓
the implication ✓

one at a time ✓
three experiments

the two theorems
open, short
a negative R_{th}
two examples

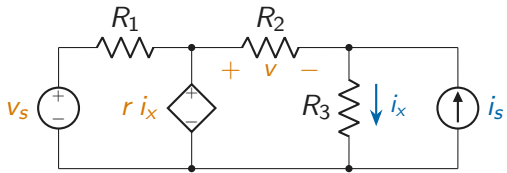
Example 13.4: three experiments



	v_s	i_s	v
Exp. 1	$12\,\mathrm{V}$	0	v_1
Exp. 2	$-6\,\mathrm{V}$	$6\,\mathrm{A}$	v_2
Exp. 3	$4\,\mathrm{V}$	$4\,\mathrm{A}$	v_3

v_3 in terms of v_1 and v_2?

Example 13.4: three experiments



	v_s	i_s	v
Exp. 1	12 V	0	v_1
Exp. 2	−6 V	6 A	v_2
Exp. 3	4 V	4 A	v_3

v_3 in terms of v_1 and v_2?

Linearity: $v = \mathbf{k} \begin{bmatrix} v_s \\ i_s \end{bmatrix}$

Example 13.4: the solution

$$\begin{bmatrix} 4 \\ 4 \end{bmatrix} = \alpha \begin{bmatrix} 12 \\ 0 \end{bmatrix} + \beta \begin{bmatrix} -6 \\ 6 \end{bmatrix}$$

Example 13.4: the solution

$$\begin{bmatrix} 4 \\ 4 \end{bmatrix} = \alpha \begin{bmatrix} 12 \\ 0 \end{bmatrix} + \beta \begin{bmatrix} -6 \\ 6 \end{bmatrix} \qquad \Rightarrow \qquad \beta = \tfrac{2}{3}, \quad \alpha = \tfrac{2}{3}$$

Example 13.4: the solution

$$\begin{bmatrix} 4 \\ 4 \end{bmatrix} = \alpha \begin{bmatrix} 12 \\ 0 \end{bmatrix} + \beta \begin{bmatrix} -6 \\ 6 \end{bmatrix} \qquad \Rightarrow \qquad \beta = \tfrac{2}{3}, \quad \alpha = \tfrac{2}{3}$$

$$v_3 = v\left(\alpha \begin{bmatrix} 12 \\ 0 \end{bmatrix} + \beta \begin{bmatrix} -6 \\ 6 \end{bmatrix}\right) = \alpha\, v_1 + \beta\, v_2$$

Example 13.4: the solution

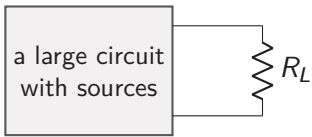
$$\begin{bmatrix} 4 \\ 4 \end{bmatrix} = \alpha \begin{bmatrix} 12 \\ 0 \end{bmatrix} + \beta \begin{bmatrix} -6 \\ 6 \end{bmatrix} \qquad \Rightarrow \qquad \beta = \tfrac{2}{3}, \quad \alpha = \tfrac{2}{3}$$

$$v_3 = v\left(\alpha \begin{bmatrix} 12 \\ 0 \end{bmatrix} + \beta \begin{bmatrix} -6 \\ 6 \end{bmatrix}\right) = \alpha\, v_1 + \beta\, v_2$$

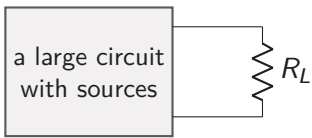
$$\boxed{v_3 = \tfrac{2}{3}\, v_1 + \tfrac{2}{3}\, v_2}$$

No element values needed

Motivation: a load

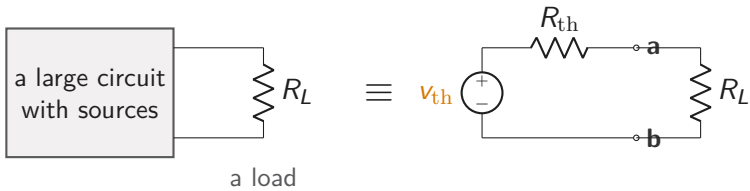


Motivation: a load



- Many loads R_L: solve the large circuit again each time?

Motivation: a load

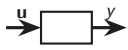


- Many loads R_L: solve the large circuit again each time?
- No: one source and one resistor, once

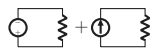
Roadmap



Input resistance



Linearity



Superposition

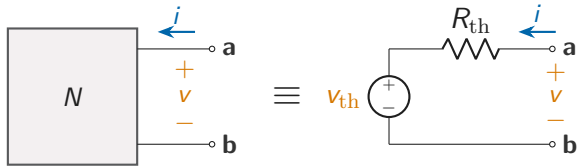


Thévenin, Norton

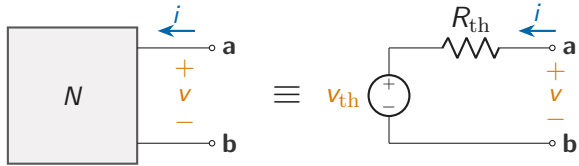


a test source ✓	$y = \mathbf{k}\,\mathbf{u}$ ✓	one at a time ✓	**the two theorems**
a dependent source ✓	the implication ✓	three experiments ✓	**open, short**
			a negative R_{th}
			two examples

Thévenin's theorem

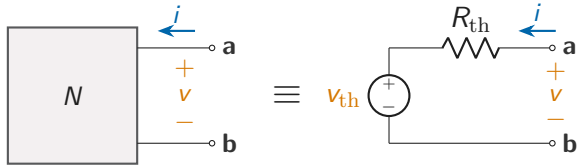


Thévenin's theorem



- N: LTI resistive, with independent sources
- (A1) Controls of the dependent sources: inside N
- (A2) The port is both voltage- and current-controlled

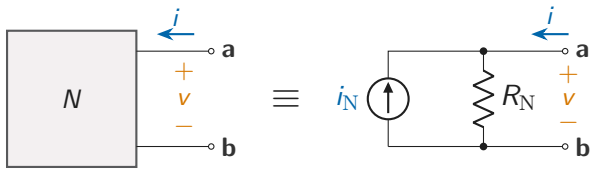
Thévenin's theorem



- N: LTI resistive, with independent sources
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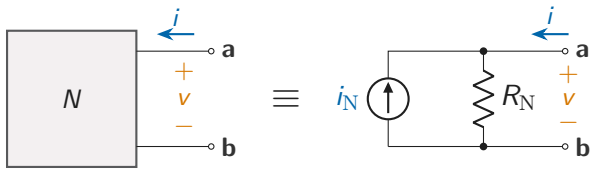
$$v = R_{\text{th}}\, i + v_{\text{th}}$$

Norton's theorem



Same conditions (A1), (A2)

Norton's theorem



Same conditions (A1), (A2)

$$v = R_N\, i + R_N\, i_N$$

Thévenin and Norton: the same line

$$v = R_{\mathrm{th}}\, i + v_{\mathrm{th}}$$

$$v = R_{\mathrm{N}}\, i + R_{\mathrm{N}}\, i_{\mathrm{N}}$$

Thévenin and Norton: the same line

$$v = R_{\mathrm{th}}\, i + v_{\mathrm{th}}$$

$$v = R_{\mathrm{N}}\, i + R_{\mathrm{N}}\, i_{\mathrm{N}}$$

$$R_{\mathrm{th}} = R_{\mathrm{N}}$$
$$v_{\mathrm{th}} = R_{\mathrm{th}}\, i_{\mathrm{N}}$$

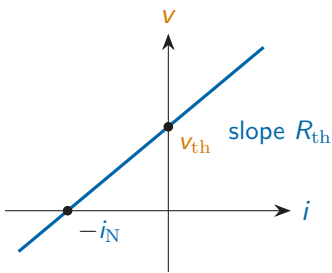
Thévenin and Norton: the same line

$v = R_{\text{th}}\, i + v_{\text{th}}$

$v = R_{\text{N}}\, i + R_{\text{N}}\, i_{\text{N}}$

$R_{\text{th}} = R_{\text{N}}$

$v_{\text{th}} = R_{\text{th}}\, i_{\text{N}}$



Thévenin and Norton: the same line

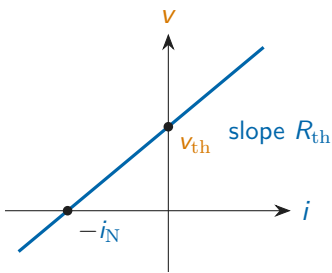
$v = R_{\text{th}}\, i + v_{\text{th}}$

$v = R_{\text{N}}\, i + R_{\text{N}}\, i_{\text{N}}$

$R_{\text{th}} = R_{\text{N}}$

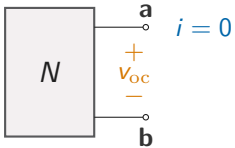
$v_{\text{th}} = R_{\text{th}}\, i_{\text{N}}$

The source transformation (previously)



Computing v_{th}, i_{N} and R_{th}

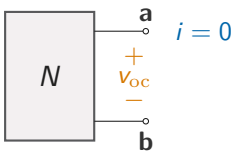
1. Open circuit



$i = 0 \;\Rightarrow\; v_{\text{oc}} = v_{\text{th}}$

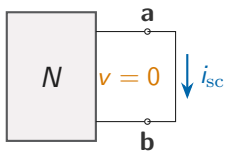
Computing v_{th}, i_{N} and R_{th}

1. Open circuit



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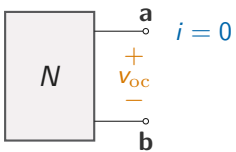
2. Short circuit



$v = 0 \;\Rightarrow\; i_{\text{sc}} = i_{\text{N}}$

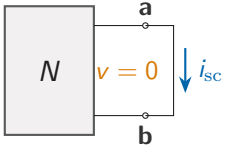
Computing v_{th}, i_{N} and R_{th}

1. Open circuit



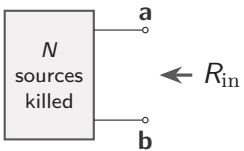
$i = 0 \;\Rightarrow\; v_{\text{oc}} = v_{\text{th}}$

2. Short circuit



$v = 0 \;\Rightarrow\; i_{\text{sc}} = i_{\text{N}}$

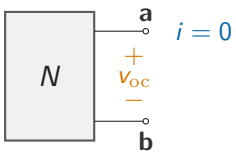
3. R_{th}



$$R_{\text{th}} = \frac{v_{\text{oc}}}{i_{\text{sc}}} \quad \text{or} \quad R_{\text{th}} = R_{\text{in}}$$

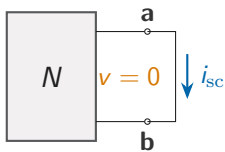
Computing v_{th}, i_{N} and R_{th}

1. Open circuit



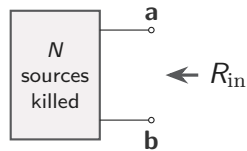
$i = 0 \;\Rightarrow\; v_{\text{oc}} = v_{\text{th}}$

2. Short circuit



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3. R_{th}



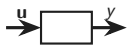
$R_{\text{th}} = \dfrac{v_{\text{oc}}}{i_{\text{sc}}}$ or $R_{\text{th}} = R_{\text{in}}$

Killing the independent sources: $v_{\text{th}} = 0$, $i_{\text{N}} = 0$, only R_{th} is left

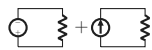
Roadmap



Input resistance



Linearity



Superposition

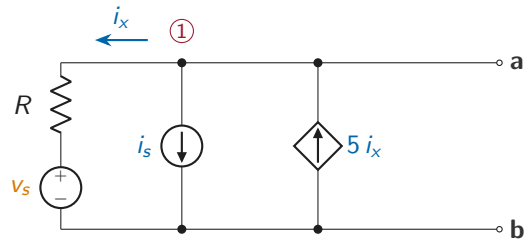


Thévenin, Norton



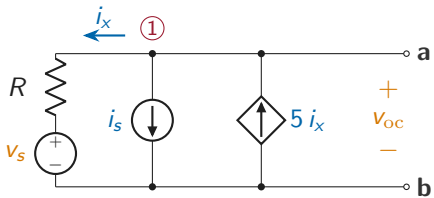
Input resistance	Linearity	Superposition	Thévenin, Norton
a test source ✓	$y = \mathbf{k}\,\mathbf{u}$ ✓	one at a time ✓	the two theorems ✓
a dependent source ✓	the implication ✓	three experiments ✓	open, short ✓
			a negative R_{th}
			two examples

Example 13.5: Thévenin and Norton equivalents



$v_{\text{th}} = ?$ $\quad i_{\text{N}} = ?$ $\quad R_{\text{th}} = ?$

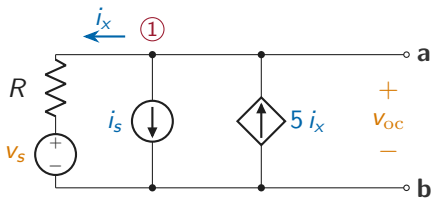
Example 13.5: step 1, open circuit



KCL at node 1:

$$i_x + i_s - 5\,i_x = 0 \;\Rightarrow\; i_x = \tfrac{1}{4}\,i_s$$

Example 13.5: step 1, open circuit



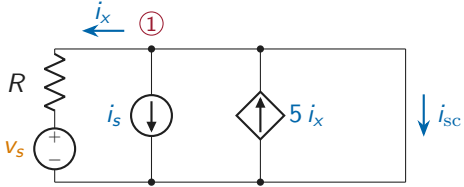
KCL at node 1:

$$i_x + i_s - 5\,i_x = 0 \;\Rightarrow\; i_x = \tfrac{1}{4}\,i_s$$

$$v_{oc} = R\,i_x + v_s$$

$$\boxed{v_{th} = \tfrac{R}{4}\,i_s + v_s}$$

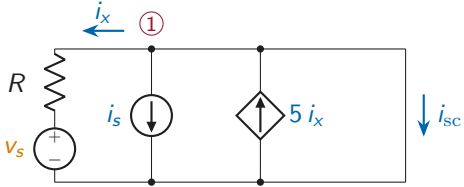
Example 13.5: step 2, short circuit



Node 1 at $0\,\mathrm{V}$:

$$R\,i_x + v_s = 0 \;\Rightarrow\; i_x = -v_s/R$$

Example 13.5: step 2, short circuit



Node 1 at $0\,\mathrm{V}$:

$R\,i_x + v_s = 0 \;\Rightarrow\; i_x = -v_s/R$

KCL at node 1:

$i_x + i_s - 5\,i_x + i_{sc} = 0$

$$i_N = i_{sc} = -i_s - \frac{4}{R}\,v_s$$

Example 13.5: step 3, R_{th}

$$R_{\text{th}} = \frac{v_{\text{th}}}{i_{\text{N}}} = \frac{\frac{R}{4}\, i_s + v_s}{-\left(i_s + \frac{4}{R}\, v_s\right)}$$

Example 13.5: step 3, R_{th}

$$R_{\text{th}} = \frac{v_{\text{th}}}{i_{\text{N}}} = \frac{\frac{R}{4}\, i_s + v_s}{-\left(i_s + \frac{4}{R}\, v_s\right)} = -\frac{R}{4}$$

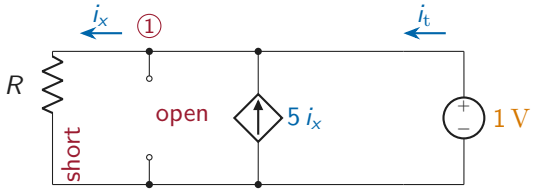
Example 13.5: step 3, R_{th}

$$R_{\text{th}} = \frac{v_{\text{th}}}{i_{\text{N}}} = \frac{\frac{R}{4}\, i_s + v_s}{-\left(i_s + \frac{4}{R}\, v_s\right)} = -\frac{R}{4}$$

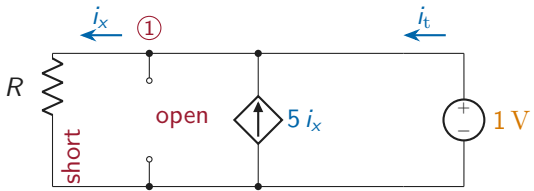
Negative: the dependent source

Example 13.5: R_{th} another way

Kill v_s and i_s; a $1\,\text{V}$ test source



Example 13.5: R_{th} another way

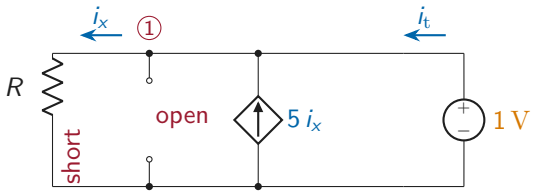


Kill v_s and i_s; a 1 V test source

$$i_x = \frac{1}{R}$$

KCL: $i_x - 5\,i_x - i_t = 0 \;\Rightarrow\; i_t = -\frac{4}{R}$

Example 13.5: R_{th} another way



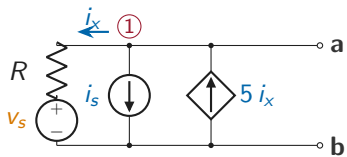
Kill v_s and i_s; a 1 V test source

$$i_x = \frac{1}{R}$$

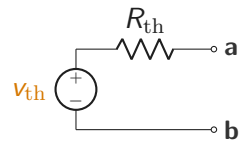
KCL: $i_x - 5\,i_x - i_t = 0 \;\Rightarrow\; i_t = -\frac{4}{R}$

$$R_{\text{th}} = R_{\text{in}} = \frac{1\,\text{V}}{i_t} = -\frac{R}{4} \quad \checkmark$$

Example 13.5: the equivalents



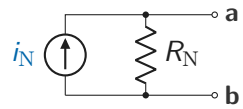
$\equiv$



$$v_{\text{th}} = \frac{R}{4}\, i_s + v_s$$

$$R_{\text{th}} = -\frac{R}{4}$$

$\equiv$



$$i_{\text{N}} = -i_s - \frac{4}{R}\, v_s$$

$$R_{\text{N}} = -\frac{R}{4}$$

Roadmap

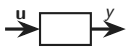


Input resistance

✓

a test source ✓

a dependent source ✓

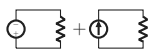


Linearity

✓

$y = \mathbf{k}\,\mathbf{u}$ ✓

the implication ✓



Superposition

✓

one at a time ✓

three experiments ✓



Thévenin, Norton

4

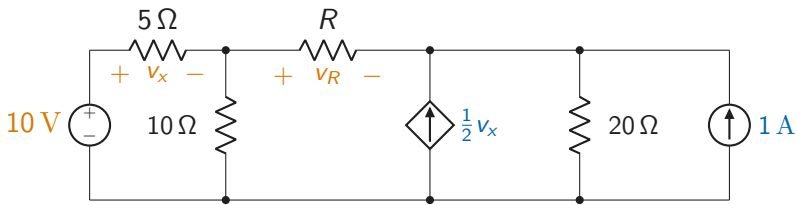
the two theorems ✓

open, short ✓

a negative R_{th} ✓

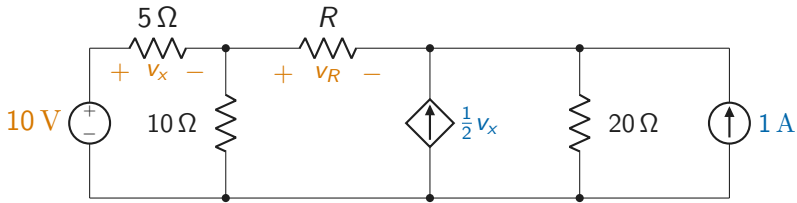
two examples

Example 13.6: a load to design



Thévenin equivalent seen by R?

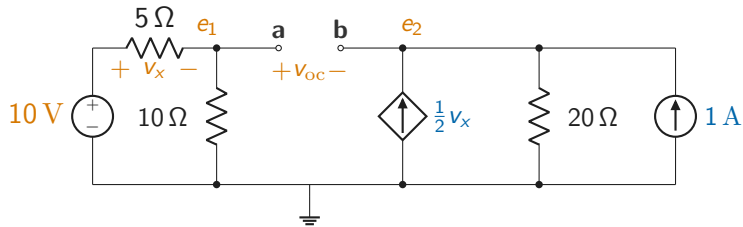
Example 13.6: a load to design



Thévenin equivalent seen by R?

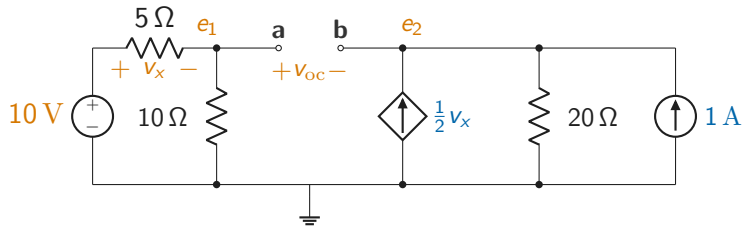
$R = ?$ for $v_R = -7\,\text{V}$

Example 13.6: open circuit



Node 1: $\dfrac{e_1 - 10}{5} + \dfrac{e_1}{10} = 0 \;\Rightarrow\; e_1 = \frac{20}{3}\,\text{V}$

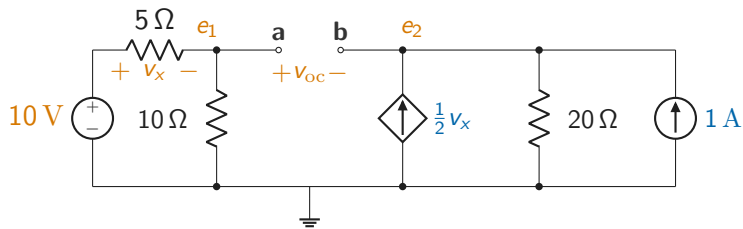
Example 13.6: open circuit



Node 1: $\dfrac{e_1 - 10}{5} + \dfrac{e_1}{10} = 0 \ \Rightarrow \ e_1 = \frac{20}{3}\ \text{V}$

$v_x = 10 - e_1 = \frac{10}{3}\ \text{V}$

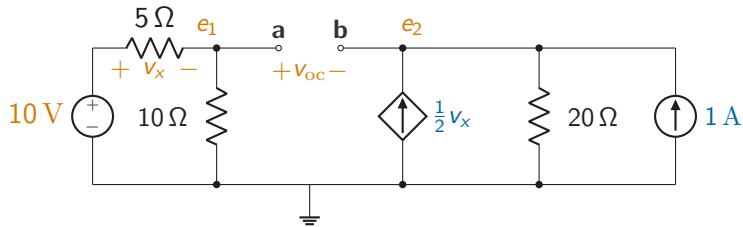
Example 13.6: open circuit



Node 1: $\dfrac{e_1 - 10}{5} + \dfrac{e_1}{10} = 0 \Rightarrow e_1 = \frac{20}{3}\,\text{V}$ **Node 2:** $-\frac{1}{2}v_x + \dfrac{e_2}{20} - 1 = 0 \Rightarrow$ $e_2 = \frac{160}{3}\,\text{V}$

$v_x = 10 - e_1 = \frac{10}{3}\,\text{V}$

Example 13.6: open circuit



Node 1: $\dfrac{e_1 - 10}{5} + \dfrac{e_1}{10} = 0 \Rightarrow e_1 = \frac{20}{3}\,\text{V}$

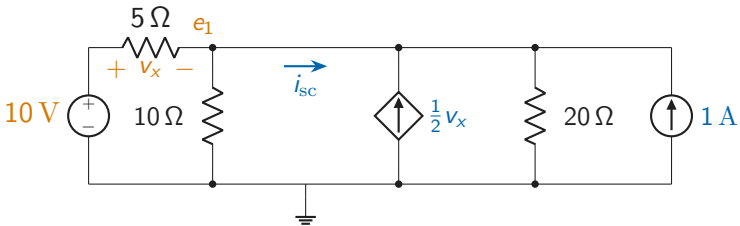
$v_x = 10 - e_1 = \frac{10}{3}\,\text{V}$

Node 2: $-\frac{1}{2}v_x + \dfrac{e_2}{20} - 1 = 0 \Rightarrow$

$e_2 = \frac{160}{3}\,\text{V}$

$$v_{\text{th}} = e_1 - e_2 = -\frac{140}{3}\,\text{V}$$

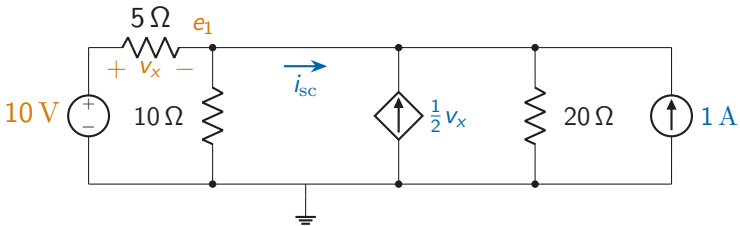
Example 13.6: short circuit



One node e_1:

$$\frac{e_1 - 10}{5} + \frac{e_1}{10} - \tfrac{1}{2}v_x + \frac{e_1}{20} - 1 = 0$$

Example 13.6: short circuit

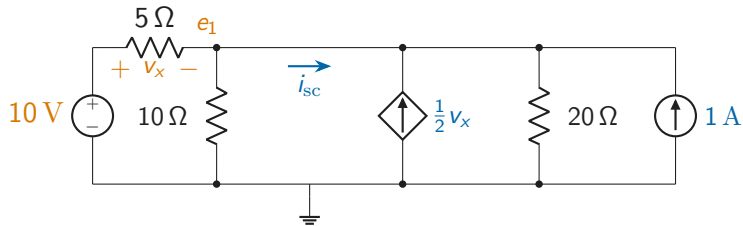


One node e_1:

$$\frac{e_1 - 10}{5} + \frac{e_1}{10} - \tfrac{1}{2}v_x + \frac{e_1}{20} - 1 = 0$$

Plug $v_x = 10 - e_1$: $\frac{17}{20}\,e_1 = 8 \;\Rightarrow\; e_1 = \frac{160}{17}\,\text{V}$

Example 13.6: short circuit



One node e_1:

$$\frac{e_1 - 10}{5} + \frac{e_1}{10} - \tfrac{1}{2}v_x + \frac{e_1}{20} - 1 = 0$$

Plug $v_x = 10 - e_1$: $\frac{17}{20}\,e_1 = 8 \;\Rightarrow\; e_1 = \frac{160}{17}\,\text{V}$

KCL, left of the short:

$$\frac{e_1 - 10}{5} + \frac{e_1}{10} + i_{sc} = 0$$

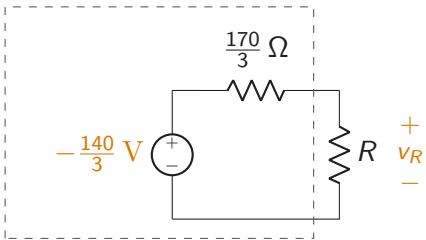
$$i_N = i_{sc} = -\tfrac{14}{17}\,\text{A}$$

Example 13.6: the load

$$R_{\text{th}} = \frac{v_{\text{th}}}{i_{\text{N}}} = \frac{-140/3}{-14/17} = \frac{170}{3}\,\Omega$$

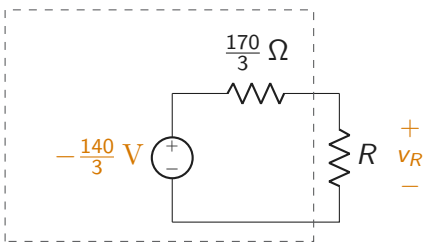
Example 13.6: the load

$$R_{\text{th}} = \frac{v_{\text{th}}}{i_{\text{N}}} = \frac{-140/3}{-14/17} = \frac{170}{3}\,\Omega$$



Example 13.6: the load

$$R_{\text{th}} = \frac{v_{\text{th}}}{i_{\text{N}}} = \frac{-140/3}{-14/17} = \frac{170}{3}\,\Omega$$

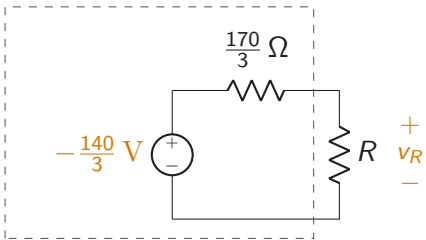


Voltage divider:

$$v_R = \frac{R}{R + \frac{170}{3}}\left(-\frac{140}{3}\right) = -7$$

Example 13.6: the load

$$R_{\text{th}} = \frac{v_{\text{th}}}{i_{\text{N}}} = \frac{-140/3}{-14/17} = \frac{170}{3}\,\Omega$$



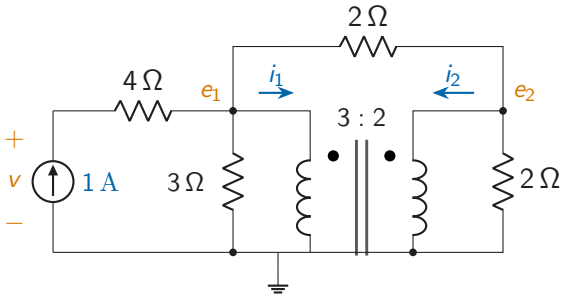
Voltage divider:

$$v_R = \frac{R}{R + \frac{170}{3}}\left(-\frac{140}{3}\right) = -7$$

$$140\,R = 21\,R + 1190$$

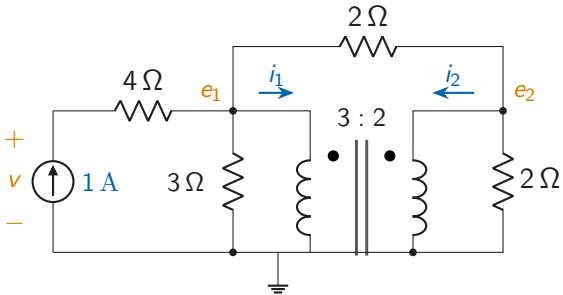
$$\boxed{R = 10\,\Omega}$$

Example 13.7: an ideal transformer



$R_{\text{in}} = ?$

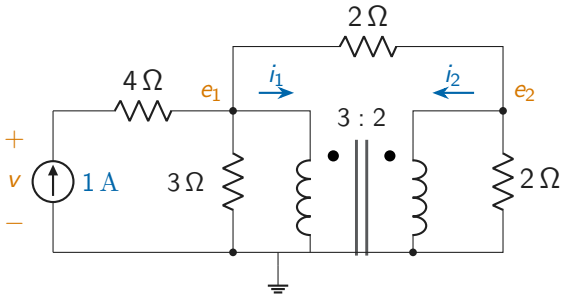
Example 13.7: an ideal transformer



$R_{\text{in}} = ?$

No independent source inside:
$R_{\text{th}} = R_{\text{in}}$

Example 13.7: an ideal transformer

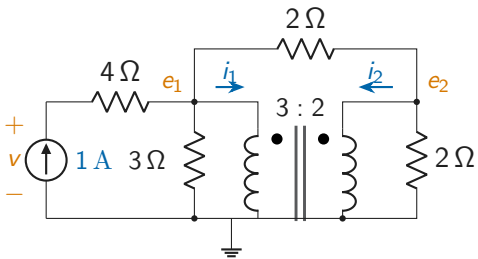


$R_{\text{in}} = ?$

No independent source inside:
$R_{\text{th}} = R_{\text{in}}$

Ideal transformer 3 : 2:
$\frac{v_1}{3} = \frac{v_2}{2}, \quad 3\,i_1 + 2\,i_2 = 0$

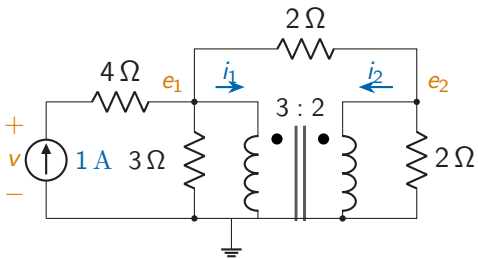
Example 13.7: the equations



Node 1: $-1 + \frac{e_1}{3} + i_1 + \frac{e_1 - e_2}{2} = 0$

Node 2: $i_2 + \frac{e_2}{2} + \frac{e_2 - e_1}{2} = 0$

Example 13.7: the equations

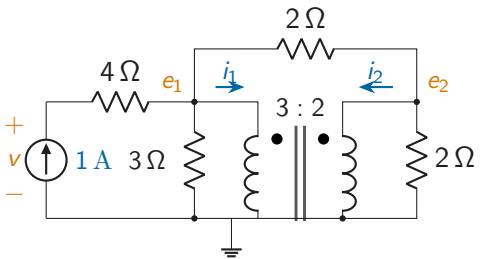


Node 1: $-1 + \dfrac{e_1}{3} + i_1 + \dfrac{e_1 - e_2}{2} = 0$

Node 2: $i_2 + \dfrac{e_2}{2} + \dfrac{e_2 - e_1}{2} = 0$

Transformer: $\dfrac{e_1}{3} = \dfrac{e_2}{2}, \quad 3\,i_1 + 2\,i_2 = 0$

Example 13.7: the equations



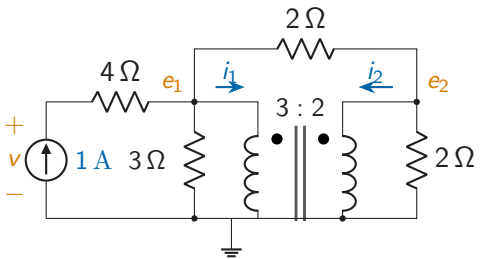
Node 1: $-1 + \dfrac{e_1}{3} + i_1 + \dfrac{e_1 - e_2}{2} = 0$

Node 2: $i_2 + \dfrac{e_2}{2} + \dfrac{e_2 - e_1}{2} = 0$

Transformer: $\dfrac{e_1}{3} = \dfrac{e_2}{2}, \quad 3\,i_1 + 2\,i_2 = 0$

4 equations, 4 unknowns: $e_1 = \frac{18}{11}\ \mathrm{V}$

Example 13.7: the equations



Node 1: $-1 + \dfrac{e_1}{3} + i_1 + \dfrac{e_1 - e_2}{2} = 0$

Node 2: $i_2 + \dfrac{e_2}{2} + \dfrac{e_2 - e_1}{2} = 0$

Transformer: $\dfrac{e_1}{3} = \dfrac{e_2}{2}, \quad 3\,i_1 + 2\,i_2 = 0$

4 equations, 4 unknowns: $e_1 = \frac{18}{11}\,\mathrm{V}$

$v = 4 \cdot 1 + e_1 = \frac{62}{11}\,\mathrm{V}$

$R_{\mathrm{in}} = \frac{62}{11}\,\Omega$

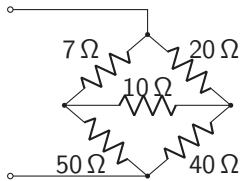
Summary

In the circuit	What to do
resistors, dependent sources	$R_{\text{in}} = v/1\,\text{A}$ with a $1\,\text{A}$ test source
linear circuit	$y = \mathbf{k}\,\mathbf{u}$: $y(\alpha\mathbf{u} + \beta\mathbf{w}) = \alpha y(\mathbf{u}) + \beta y(\mathbf{w})$
several sources	superposition: killed v_s = short, killed i_s = open
independent sources inside	$v_{\text{th}} = v_{\text{oc}}$, $i_{\text{N}} = i_{\text{sc}}$, $R_{\text{th}} = R_{\text{N}} = v_{\text{oc}}/i_{\text{sc}}$ or R_{in} with the sources killed

Dependent sources are never killed; R_{th} may be negative

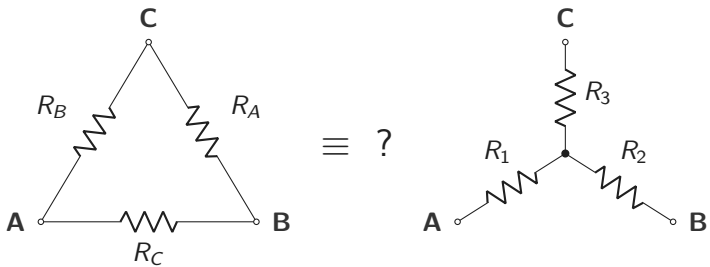
Take-home

The Δ–Y transform

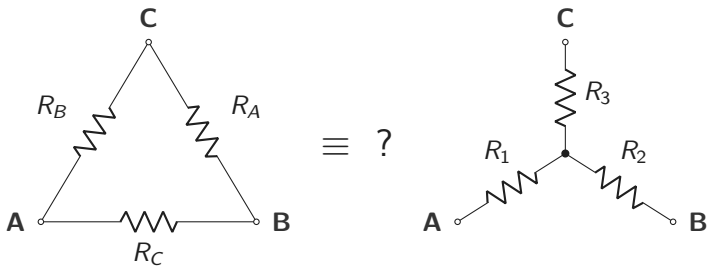


No two resistors in series or in parallel

Three-terminal circuits: Δ and Y

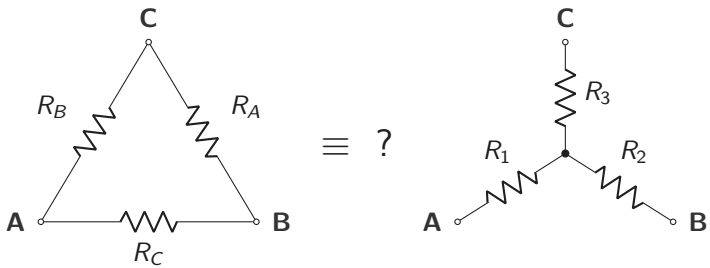


Three-terminal circuits: Δ and Y



Given R_A, R_B, R_C: find R_1, R_2, R_3

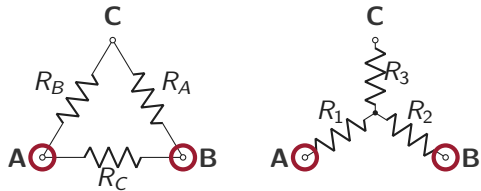
Three-terminal circuits: Δ and Y



Given R_A, R_B, R_C: find R_1, R_2, R_3

The same resistance between every two terminals

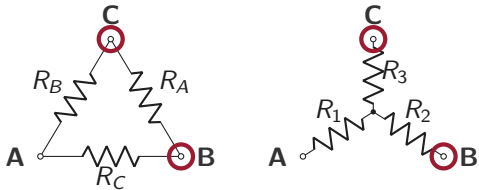
Δ to Y: the resistance between two terminals



Δ Y

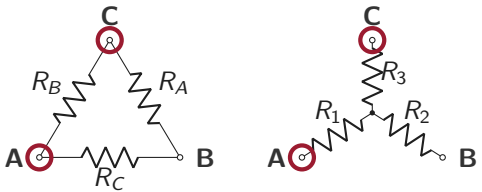
A–B: $$\frac{R_C(R_A + R_B)}{R_A + R_B + R_C} = R_1 + R_2 \quad (1)$$

Δ to Y: the resistance between two terminals



	Δ		Y	
A–B:	$\dfrac{R_C(R_A+R_B)}{R_A+R_B+R_C}$	$=$	R_1+R_2	(1)
B–C:	$\dfrac{R_A(R_B+R_C)}{R_A+R_B+R_C}$	$=$	R_2+R_3	(2)

Δ to Y: the resistance between two terminals



	Δ		Y	
A–B:	$\dfrac{R_C(R_A+R_B)}{R_A+R_B+R_C}$	$=$	R_1+R_2	(1)
B–C:	$\dfrac{R_A(R_B+R_C)}{R_A+R_B+R_C}$	$=$	R_2+R_3	(2)
C–A:	$\dfrac{R_B(R_C+R_A)}{R_A+R_B+R_C}$	$=$	R_3+R_1	(3)

Δ to Y: the result

$$\frac{1}{2}\left[(1)+(2)+(3)\right]: \quad R_1+R_2+R_3=\frac{R_AR_B+R_BR_C+R_CR_A}{R_A+R_B+R_C} \tag{4}$$

Δ to Y: the result

$$\tfrac{1}{2}\left[(1)+(2)+(3)\right]: \quad R_1+R_2+R_3=\frac{R_AR_B+R_BR_C+R_CR_A}{R_A+R_B+R_C} \quad (4)$$

$$R_1=\frac{R_BR_C}{R_A+R_B+R_C} \quad (4)-(2)$$

Δ to Y: the result

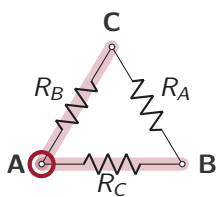
$$\tfrac{1}{2}\big[(1)+(2)+(3)\big]: \quad R_1+R_2+R_3=\frac{R_AR_B+R_BR_C+R_CR_A}{R_A+R_B+R_C} \quad (4)$$

$$R_1=\frac{R_BR_C}{R_A+R_B+R_C} \quad (4)-(2) \qquad R_2=\frac{R_CR_A}{R_A+R_B+R_C} \qquad R_3=\frac{R_AR_B}{R_A+R_B+R_C}$$

Δ to Y: the result

$$\tfrac{1}{2}[(1)+(2)+(3)]: \quad R_1+R_2+R_3=\frac{R_AR_B+R_BR_C+R_CR_A}{R_A+R_B+R_C} \quad (4)$$

$$R_1=\frac{R_BR_C}{R_A+R_B+R_C} \quad (4)-(2) \qquad R_2=\frac{R_CR_A}{R_A+R_B+R_C} \qquad R_3=\frac{R_AR_B}{R_A+R_B+R_C}$$



Y resistor at a terminal = product of the two Δ resistors there / sum of the Δ resistors

Y to Δ

Given R_1, R_2, R_3: find R_A, R_B, R_C

Y to Δ

Given R_1, R_2, R_3: find R_A, R_B, R_C

$$R_1R_2 + R_2R_3 + R_3R_1 = \frac{R_AR_BR_C\,(R_A + R_B + R_C)}{(R_A + R_B + R_C)^2} = \frac{R_AR_BR_C}{R_A + R_B + R_C}$$

Y to Δ

Given R_1, R_2, R_3: find R_A, R_B, R_C

$$R_1R_2 + R_2R_3 + R_3R_1 = \frac{R_AR_BR_C\,(R_A + R_B + R_C)}{(R_A + R_B + R_C)^2} = \frac{R_AR_BR_C}{R_A + R_B + R_C}$$

$$R_AR_1 = R_BR_2 = R_CR_3 = \frac{R_AR_BR_C}{R_A + R_B + R_C}$$

Y to Δ

Given R_1, R_2, R_3: find R_A, R_B, R_C

$$R_1R_2 + R_2R_3 + R_3R_1 = \frac{R_AR_BR_C\,(R_A + R_B + R_C)}{(R_A + R_B + R_C)^2} = \frac{R_AR_BR_C}{R_A + R_B + R_C}$$

$$R_AR_1 = R_BR_2 = R_CR_3 = \frac{R_AR_BR_C}{R_A + R_B + R_C}$$

$$R_A = \frac{R_1R_2 + R_2R_3 + R_3R_1}{R_1}, \qquad R_B = \frac{\cdots}{R_2}, \qquad R_C = \frac{\cdots}{R_3}$$

Y to Δ

Given R_1, R_2, R_3: find R_A, R_B, R_C

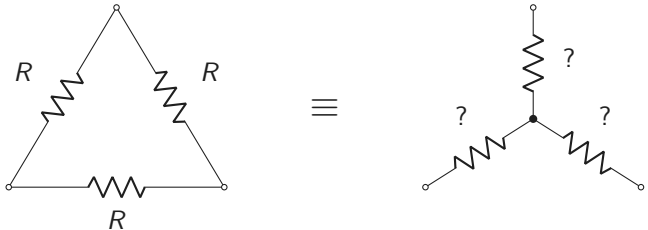
$$R_1R_2 + R_2R_3 + R_3R_1 = \frac{R_AR_BR_C\,(R_A + R_B + R_C)}{(R_A + R_B + R_C)^2} = \frac{R_AR_BR_C}{R_A + R_B + R_C}$$

$$R_AR_1 = R_BR_2 = R_CR_3 = \frac{R_AR_BR_C}{R_A + R_B + R_C}$$

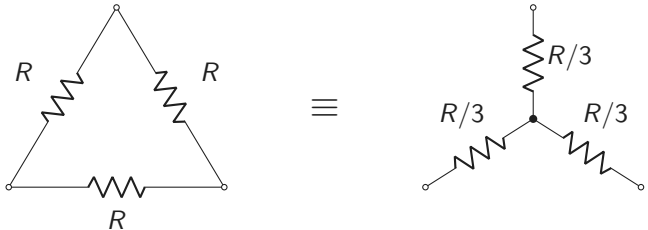
$$R_A = \frac{R_1R_2 + R_2R_3 + R_3R_1}{R_1}, \qquad R_B = \frac{\cdots}{R_2}, \qquad R_C = \frac{\cdots}{R_3}$$

$$\Delta \text{ resistor} = \frac{\text{sum of the products of the Y resistors}}{\text{the Y resistor opposite}}$$

Equal resistors

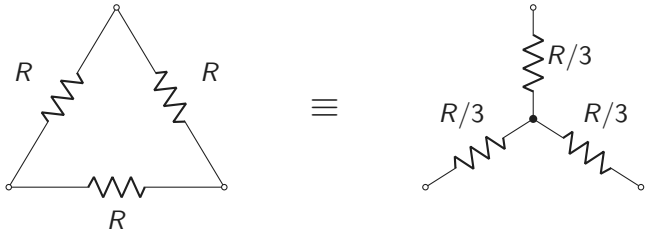


Equal resistors



$$\frac{R \cdot R}{3R} = \frac{R}{3}$$

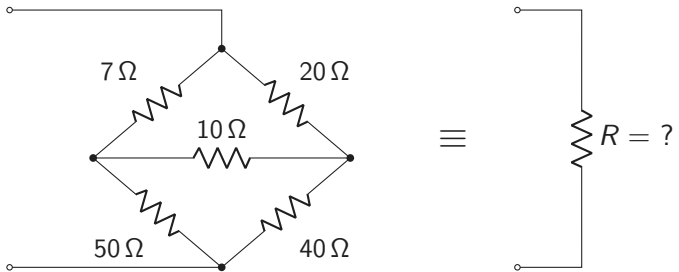
Equal resistors



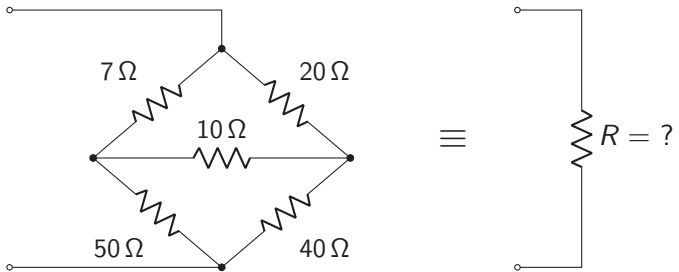
$$\frac{R \cdot R}{3R} = \frac{R}{3}$$

Y with $R \equiv \Delta$ with $3R$

Example 13.8: a bridge

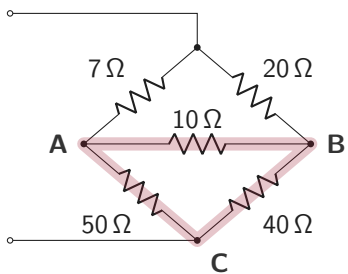


Example 13.8: a bridge



No two resistors in series or in parallel

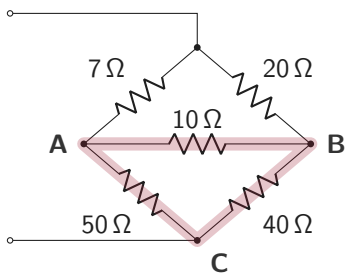
Example 13.8: Δ to Y



Δ A–B–C:

$R_C = 10\,\Omega, \quad R_B = 50\,\Omega, \quad R_A = 40\,\Omega$

Example 13.8: Δ to Y

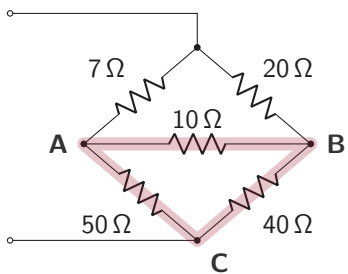


Δ A–B–C:

$R_C = 10\,\Omega, \quad R_B = 50\,\Omega, \quad R_A = 40\,\Omega$

$R_A + R_B + R_C = 100\,\Omega$

Example 13.8: Δ to Y



Δ A–B–C:

$R_C = 10\,\Omega, \quad R_B = 50\,\Omega, \quad R_A = 40\,\Omega$

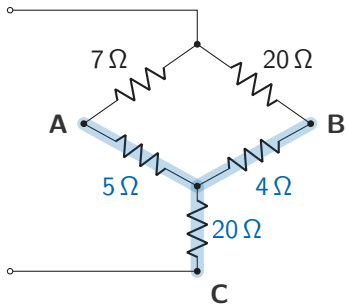
$R_A + R_B + R_C = 100\,\Omega$

$$R_1 = \frac{50 \cdot 10}{100} = 5\,\Omega$$

$$R_2 = \frac{10 \cdot 40}{100} = 4\,\Omega$$

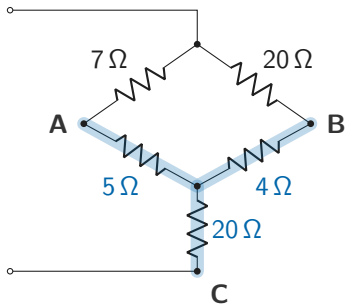
$$R_3 = \frac{40 \cdot 50}{100} = 20\,\Omega$$

Example 13.8: the new circuit



Series: $7 + 5 = 12\,\Omega, \quad 20 + 4 = 24\,\Omega$

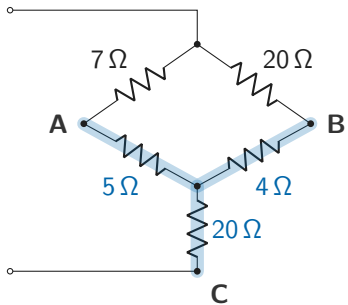
Example 13.8: the new circuit



Series: $7 + 5 = 12\,\Omega, \quad 20 + 4 = 24\,\Omega$

Parallel: $12 \parallel 24 = 8\,\Omega$

Example 13.8: the new circuit



Series: $7 + 5 = 12\,\Omega, \quad 20 + 4 = 24\,\Omega$

Parallel: $12 \parallel 24 = 8\,\Omega$

Series: $8 + 20$

$$R = 28\,\Omega$$